2) User Stories & Use Cases

Multi-Platform Shorts Automation & AI Insights Engine

This section describes how the main users interact with the system to automate short-form video publishing and receive AI-generated performance insights. The scenarios are based on the project concept and workflow.

1. User Stories

User stories express the system's functionality from the perspective of the people who use or benefit from it.

US-01 — Automated Multi-Platform Publishing

As a content creator, I want to upload a short video to a designated Google Drive folder so that the system can publish it to YouTube Shorts, Instagram Reels, and Facebook Reels without requiring me to upload it manually to each platform.

US-02 — Centralized Video Metadata

As a content creator, I want the system to retrieve the video's metadata and associated caption information so that the correct details can be used during publishing and recorded centrally.

US-03 — Publishing Status Tracking

As a content creator, I want the system to record each video's publishing status in Google Sheets and move successfully processed files to the Published_Shorts folder so that I can monitor completed publishing tasks.

US-04 — Automatic Performance Collection

As a digital marketer, I want the system to collect performance metrics 24 to 72 hours after publication so that I can evaluate how each video performs across supported platforms.

US-05 — AI-Powered Performance Analysis

As a content creator, I want the system to analyze views, likes, watch time or retention, shares, and comments so that I can understand engagement patterns and identify opportunities to improve future videos.

US-06 — Actionable Optimization Recommendations

As a digital marketer, I want to receive concise AI-generated recommendations about hooks, captions, hashtags, calls to action, and posting times so that I can make data-informed content decisions.

US-07 — Insight Delivery and Storage

As a content creator, I want the AI report to be delivered through Telegram or email and saved in Google Sheets so that I can access recommendations and compare insights over time.

2. Use Cases

Use cases describe a user goal, the system's response, and the expected outcome in a specific scenario.

UC-01 — Publish a Short Video Across Platforms

Primary actor: Content Creator

Preconditions: The creator has access to the designated Google Drive folder; the workflow is configured; and required platform credentials and APIs are available.

Trigger: A new .mp4 video is added to the Pending Shorts folder.

Main flow:

The n8n trigger detects the new video.

The workflow retrieves the relevant metadata from the configured source, including Google Sheets when applicable.

The workflow prepares the media and platform-specific publishing requests.

The system sends publishing requests to the supported platforms: YouTube, Instagram, and Facebook.

The workflow records the publishing status and updates the file organization.

Expected outcome: The video is submitted for publishing across the configured platforms, and its status is logged. Actual success depends on platform API permissions, supported formats, and each platform's response.

Alternative / exception flow: If a platform request fails, the workflow should record the failure for review. Automatic retries or partial-failure handling must be configured if required.

UC-02 — Review Publishing Status

Primary actor: Content Creator

Preconditions: Publishing status logging is enabled in Google Sheets.

Trigger: The creator wants to check whether a video has been processed.

Main flow:

The creator opens the project's Google Sheets tracking log.

The creator locates the video record.

The creator reviews the recorded status and any available processing details.

Expected outcome: The creator can identify the recorded publishing status without checking every platform manually.

Alternative / exception flow: If the record is missing or incomplete, the creator reviews the workflow execution log and corrects the tracking data as needed.

UC-03 — Collect Video Performance Metrics

Primary actor: System (scheduled workflow), on behalf of the Content Creator

Preconditions: The video has been published, the scheduled trigger is configured, and the required analytics APIs and permissions are available.

Trigger: The scheduled workflow reaches the configured post-publication interval, such as 24 or 72 hours.

Main flow:

The schedule trigger starts the analytics workflow.

The system requests available metrics from the relevant platform APIs.

The workflow associates the returned metrics with the correct video and platform.

The system stores the collected metrics in the centralized tracking sheet.

Expected outcome: Available performance data is organized for later analysis. The exact metrics depend on the permissions and data exposed by each platform API.

Alternative / exception flow: If metrics are unavailable or an API request fails, the workflow records the issue; later retries require an appropriate retry policy.

UC-04 — Generate AI Performance Insights

Primary actor: Content Creator or Digital Marketer

Preconditions: Performance metrics have been collected and are available to the AI analysis workflow; the AI provider API is configured.

Trigger: The analytics workflow submits the collected metrics for analysis.

Main flow:

The system sends the relevant metrics and context to the configured AI model.

The AI analyzes engagement patterns, such as views, likes, shares, watch time or retention, and comments where available.

The AI generates concise findings and recommendations about hooks, captions, hashtags, calls to action, and posting times when supported by the data.

The workflow formats the analysis into a report.

Expected outcome: The user receives an AI-generated summary and practical suggestions for improving future content. Recommendations are based on the available data and should not be treated as guaranteed outcomes.

Alternative / exception flow: If the AI response is invalid or incomplete, validation and retry handling should be configured before the report is delivered.

UC-05 — Receive and Review the AI Report

Primary actor: Content Creator or Digital Marketer

Preconditions: The report has been generated and Telegram or email delivery is configured.

Trigger: The analysis workflow completes report generation.

Main flow:

The workflow sends the report through the configured notification channel.

The workflow saves the report or its key insights in Google Sheets.

The user reviews the findings and selects recommendations to test in upcoming videos.

Expected outcome: The user can access the report and use the findings to guide future content decisions.

Alternative / exception flow: If notification delivery fails, the workflow should log the failure; an alternative channel or retry can be used if configured.

3. Representative End-to-End Scenario

A content creator uploads an MP4 file to the Pending_Shorts folder in Google Drive. The n8n workflow detects the file, retrieves the configured metadata, prepares platform-specific publishing requests, and records the resulting status in Google Sheets. After the configured 24- or 72-hour interval, another workflow collects available analytics metrics and sends them to the AI analysis step. The system generates a concise report, delivers it through Telegram or email, and stores the insights in Google Sheets. The creator reviews the recommendations and applies suitable changes to future videos.

4. Assumptions and Dependencies

The project owner configures and maintains the required API credentials, permissions, and platform integrations.

Publishing and analytics capabilities depend on each platform's current API access, account type, policies, and supported features.

The project overview describes the intended workflow; retry behavior, detailed error handling, duplicate detection, and success criteria must be defined during implementation.

The system can only analyze metrics that are returned by the connected APIs and stored correctly.